

Day 4 (03-11-2025)

⇒ Write a Program to calculate square of a given number

```
n = int(input())
```

```
print(n*n)
```

15 ⇒ o/p ⇒ 225

⇒ a = 5
print(a*a) ⇒ type(a)
 o/p - int
 ⇒ 25

⇒ ~~a = int(input())~~ ⇒ a = input()
~~print(a)~~ type(a)
type o/p ⇒ string
o/p -

the user take input defaultly as a string. Suppose we to make it int we have to do this.

```
a = int(input())
```

```
print(a*a)
```

6 o/p ⇒ 36

Data types -

A data is classification that tells a computer what kind of value a variable holds & what operations can be performed on what data.

(or).

A data type define the type of a data a variable can store such as numbers, text or logical values.

Single value data types - It is also called primitive data types or scalar data types.
This data type that can hold only one value at a time - not collections or multiple values like arrays or objects.

- int
- float
- complex
- boolean

Multi value data types - It is also called non-primitive or composite data types)

This data type can hold (or) store multiple values at once - unlike single value data types that hold only one.

- str
- list
- tuple
- set
- dictionary

It is also called as data structures

⇒ $a = 23.8$
type(a)
⇒ float
o/p - Complex

⇒ $a = 2 + 3j$
type(a)

o/p - Complex

⇒ $a = \text{True}$
type(a)
o/p - bool

⇒ $a = 2, "pooyy", 8.6, \text{false}$
type(a)
o/p - tuple

Declaring the inputs -

1) $a = 6$
 $b = 7$
 $c = 8$
print(a, b, c)

o/p → 6 7 8

2) $x = 3; y = 4; z = 5$
print(x, y, z)

o/p → 3 4 5

3) $m, n, p = 10, 20, 30$
print(m, n, p)

o/p - 10, 20, 30

Program

Write a program to swapping the number

o/p

before swap

a=10 and b=20

after swap

a=20 and b=10

try with the 3rd variable & without 3rd variable.

⇒ # 3rd variable method

a=10

b=20

print(f" before swap a={a} and b={b}")

temp=a # 3rd variable taken as temp & store a value.

a=b # now b value to a

b=temp # at last temp value to b.

print(f" after swap a={a} and b={b}")

o/p) before swap a=10 and b=20

after swap a=20 and b=10

⇒ Without 3rd variable method ②

a=10

b=20

print(f" before swap a={a} and b={b}")

~~a,b~~

a,b = b,a

print(f" after swap a={a} and b={b}")

o/p) before swap a=10 and b=20

after swap a=20 and b=10

Program

2) Write a Program to calculate area of rectangle, triangle, circle?

'''
 $aor = l \times b$

$aot = 0.5 \times \text{base} \times \text{height}$

$aoc = 3.142 \times r \times r$ '''

$l = \text{float}(\text{input}(\text{"enter a len of rec:"}))$

$b = \text{float}(\text{input}(\text{"enter a breadth of rec:"}))$

$\text{base} = \text{float}(\text{input}(\text{"enter a base of tri:"}))$

$\text{height} = \text{float}(\text{input}(\text{"enter a height of tri:"}))$

$r = \text{float}(\text{input}(\text{"enter a radius of circle:"}))$

$aor = l * b$

$\text{print}(\text{"the area of rectangle = {aor} \n the area of triangle = {0.5 * base * height} \n the area of circle = {3.142 * r * r}"})$

Taking i/p ->

enter a len of rec: 5

enter a breadth of rec: 3.5

enter a base of tri: 6

enter a height of tri: 9

enter a radius of circle: 2.9

dp) the area of rec = 17.5

the area of tri = 27.0

the area of circle = 26.424219999999995

Day 7 (07/11/2025)

Nested-If statement -

if statement can also be checked inside other if statement. This conditional statement is called a nested if statement. This means inner if condition will only be checked if the outer if condition is True.

Syntax of nested if else -

```
if (condition1): # outer if
    if (condition2): # inner if
        statements of inner if
    else: # inner else
        statements of inner else
```

```
else:
    statements of outer else.
```

```
Q) age = int(input())
country = input()
if country == "india":
    if (age >= 18):
        print("You are eligible for voting")
```

```
else:
```

```
    print("not eligible for voting bcz your age is not more than 17")
```

```
else:
    print("not eligible for voting bcz your country is not valid")
```

i/p $\Rightarrow$ 19
china

o/p $\Rightarrow$ not eligible for voting bcz your country is not valid.

```

a) age = int(input())
country = input()
if (country == "india" or age >= 18):
    if country == "india":
        if (age >= 18):
            print("you are eligible for voting")
        else:
            print("not eligible for voting bcz your age
            is not more than 18.")
    else:
        print("not eligible for voting bcz your country
        is not valid")
else:
    print("both age & country are invalid")

```

i/p => 11 o/p => both age & country are invalid.
 Japan.

```

b) # using nested if else point number is +ve, -ve, zero.
num = int(input())
if num >= 0:
    if num == 0:
        print("zero")
    else:
        print("+ve")
else:
    print("-ve")

```

i/p => 8 o/p => +ve

Q) Triangle type checker

take three sides of triangle as input

and check it is equilateral, isosceles, scalene or not a triangle.

~~side~~

```
a = int(input())
```

```
b = int(input())
```

```
c = int(input())
```

```
if (a+b>c) and (a+c>b) and (b+c>a):
```

```
    if (a==b==c):
```

```
        print("equilateral triangle")
```

```
    elif (a==b or b==c or c==a):
```

```
        print("isosceles triangle")
```

```
    else:
```

```
        print("scalene triangle")
```

```
else:
```

```
    print("It's not a triangle")
```

```
i/p => 1
```

```
1
```

```
3
```

```
o/p => It's not a triangle.
```